Chapter 1. Time Series Data

Dependence, stationarity and autocovariance

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1.1 Time series data with dependence

Motivation. Introductory statistics rests on the i.i.d. assumption, but observations recorded over time (temperatures, stock prices, economic indicators) are typically correlated with their own past. Time series analysis studies such dependent data; this section shows the simplest descriptive devices (smoothing) and explains why a model for the dependence is needed.

1.1.1 Trend and moving-average smoothing

Let $x_1, \dots, x_n$ be observations, e.g. monthly closing values of a stock index. Raw prices fluctuate so much that the underlying movement is hard to see. The *moving average* of span m,

$$y_t = \frac{x_{t-m+1} + \cdots + x_t}{m},$$

replaces each value by the mean of the last m observations. Larger m gives a smoother curve; a short-span average describes short-run movement, a long-span one the long-run trend. Producing such a curve is called *smoothing*.

To see why it works, suppose the data follow the signal-plus-noise model

$$x_t = f(t) + z_t, \qquad t = 1, \dots, n, \tag{1.1}$$

where f is an unknown deterministic function and z_t are i.i.d. errors with mean 0 and variance σ^2. Then

$$y_t = \frac{f(t-m+1) + \cdots + f(t)}{m} + \frac{z_{t-m+1} + \cdots + z_t}{m}. \tag{1.2}$$

For moderately large m the second term is close to $Ez_1 = 0$ by the law of large numbers (its variance is σ^2/m), so $y_t \approx f_m(t) := m^{-1} \sum_{j=0}^{m-1} f(t-j)$, a local average of the trend.

1.1.2 Exponential smoothing

An alternative to equal weights is to discount the past geometrically. For $0 < w < 1$ put

$$S_t = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j}.$$

Given an initial value S_0^*, define recursively

$$S_t^* = w x_t + (1-w) S_{t-1}^*. \tag{1.3}$$

Iterating (1.3) gives

$$S_t^* = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j} + (1-w)^t S_0^* = S_t + (1-w)^t S_0^*, \tag{1.4}$$

so $S_t^* \approx S_t$ for large t, and the cheap recursion may be used instead of S_t. Common choices of S_0^* are x_1 or the sample mean; w can be chosen to minimize the one-step prediction error

$$SSE(w) = \sum_{t=2}^{n} (x_t - S_{t-1}^*)^2. \tag{1.5}$$

1.1.3 Trend, seasonality and noise

Writing T_t for the trend $f(t)$, one also uses the decomposition $x_t = T_t + S_t + z_t$, where the *seasonal component* satisfies $S_t = S_{t+s}$ for a period s ($s = 4$ for quarterly, $s = 12$ for monthly data). The i.i.d. assumption on z_t is only a convenience: real errors are usually strongly correlated. The next section introduces stationarity, the basic framework for modelling such dependence.

1.2 Stationary time series

Motivation. To do inference from a single realization we need the probabilistic structure to be stable over time. Stationarity formalizes this, and the autocovariance function becomes the central object describing the dependence.

Definition 1.2.1 (Time series)

If T is the set of integers $\mathbb{Z}$ or of positive integers $\mathbb{Z}^+ = \{1, 2, \dots\}$, a collection of random variables $\{X_t : t \in T\}$ is called a *time series*.

Definition 1.2.2 (Strictly stationary time series)

$\{X_t\}$ is *strictly stationary* if for all $k \geq 1$, all integers $t_1, \dots, t_k$ and h, and all real $x_1, \dots, x_k$,

$$P(X_{t_1} \leq x_1, \dots, X_{t_k} \leq x_k) = P(X_{t_1+h} \leq x_1, \dots, X_{t_k+h} \leq x_k).$$

That is, every finite-dimensional distribution is invariant under time shifts.

Definition 1.2.3 (Weakly stationary time series)

$\{X_t\}$ is *(weakly) stationary* if
(1) $EX_t = \mu$ for all t;
(2) $EX_t^2 < \infty$ for all t;
(3) $\mathrm{Cov}(X_t, X_{t+h}) = \mathrm{Cov}(X_0, X_h)$ for all t, h.

Condition (1) says the level is constant, (2) that variances exist, and (3), the essential one, that the covariance between two observations depends only on their time distance; in particular the variance is constant. Unless stated otherwise, "stationary" means weakly stationary.

Example 1.2.4 (Strict versus weak stationarity)

(1) An i.i.d. sequence $\{X_t\}$ is strictly stationary.
(2) If moreover $EX_1^2 < \infty$, it is also weakly stationary. An i.i.d. Cauchy sequence is strictly but not weakly stationary, since the variance does not exist.
(3) If ϵ_t are i.i.d. with mean 0, variance 1, then $X_t = \epsilon_t + \epsilon_{t-1}$ is strictly and weakly stationary.

Neither notion contains the other, but a strictly stationary series with finite variance is weakly stationary: shift invariance of the bivariate law of (X_t, X_{t+h}) gives (1) and (3). Throughout, stationary series are assumed to have finite variance.

Definition 1.2.5 (Autocovariance function)

For a stationary $\{X_t\}$ the real function

$$\gamma(h) = \mathrm{Cov}(X_t, X_{t+h}), \qquad h = 0, \pm 1, \pm 2, \ldots,$$

is called the *autocovariance function* (ACVF).

Theorem 1.2.6 (Basic properties of the ACVF)

The autocovariance function satisfies
(1) $\gamma(0) \geq 0$;
(2) $\gamma(-h) = \gamma(h)$;
(3) $|\gamma(h)| \leq \gamma(0)$.

Proof. [Supplement]

Step 1. $\gamma(0) = \mathrm{Var}(X_t) \geq 0$.

Step 2. By stationarity and symmetry of covariance, $\gamma(-h) = \mathrm{Cov}(X_t, X_{t-h}) = \mathrm{Cov}(X_{t-h}, X_t) = \gamma(h)$, the last equality by putting $s = t - h$.

Step 3. By the Cauchy–Schwarz inequality, $|\mathrm{Cov}(X_t, X_{t+h})| \leq \sqrt{\mathrm{Var}\, X_t \ \mathrm{Var}\, X_{t+h}} = \gamma(0)$. ■

Theorem 1.2.7 (Nonnegative definiteness)

For any reals $x_1, \ldots, x_k$ and integers $t_1, \ldots, t_k$,

$$\sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j \geq 0.$$

Proof. [Supplement] Let $Y = \sum_{i=1}^{k} x_i X_{t_i}$. Since $\mathrm{Cov}(X_{t_i}, X_{t_j}) = \gamma(t_j - t_i) = \gamma(t_i - t_j)$ by Theorem 1.2.6(2),

$$0 \leq \mathrm{Var}(Y) = \sum_{i,j=1}^{k} x_i x_j \,\mathrm{Cov}(X_{t_i}, X_{t_j}) = \sum_{i,j=1}^{k} x_i \gamma(t_i - t_j) x_j.$$

■

In matrix form: the covariance matrix $\Gamma_k = [\gamma(t_i - t_j)]_{i,j=1}^{k}$ of any finite sub-vector is nonnegative definite. (Conversely, every even nonnegative definite function on $\mathbb{Z}$ is the ACVF of some stationary, indeed Gaussian, series.)

Definition 1.2.8 (White noise)

A stationary series $\{X_t\}$ whose ACVF satisfies $\gamma(h) = 0$ for $h \neq 0$ is called *white noise*, written $\{X_t\} \sim WN(0, \sigma^2)$ with $\sigma^2 = \gamma(0)$ (mean zero is understood). If in addition the X_t are i.i.d., we write $\{X_t\} \sim IID(0, \sigma^2)$.

Definition 1.2.9 (Autocorrelation function)

For a stationary $\{X_t\}$,

$$\rho(h) = \gamma(h)/\gamma(0)$$

is the *autocorrelation function* (ACF); for fixed h, $\rho(h)$ is the autocorrelation coefficient at lag h.

By Theorem 1.2.6, $\rho(0) = 1$, $\rho(-h) = \rho(h)$ and $|\rho(h)| \le 1$.

Definition 1.2.10 (Multivariate normal distribution)

Let $\mu = (\mu_1, \dots, \mu_k)^T \in \mathbb{R}^k$ and let $\Sigma = (\sigma_{ij})_{i,j=1}^k$ be symmetric positive definite: $\sigma_{ij} = \sigma_{ji}$ and $a^T \Sigma a > 0$ for all $a \in \mathbb{R}^k \setminus \{0\}$. If the random vector $Z = (Z_1, \dots, Z_k)^T$ has joint density

$$f(z) = \frac{1}{(2\pi)^{k/2}|\Sigma|^{1/2}} \exp\Big\{-\tfrac{1}{2}(z-\mu)^T \Sigma^{-1}(z-\mu)\Big\}, \qquad z \in \mathbb{R}^k,$$

then Z follows the *multivariate normal distribution* with mean μ and covariance Σ; here $\mu_i = EZ_i$ and $\sigma_{ij} = \mathrm{Cov}(Z_i, Z_j)$.

Definition 1.2.11 (Gaussian time series)

$\{X_t\}$ is a *Gaussian time series* if $(X_{t_1}, \dots, X_{t_k})^T$ is multivariate normal for every k and all $t_1, \dots, t_k$.

Since a normal law is determined by its mean and covariance, a weakly stationary Gaussian series is strictly stationary. Normality rarely holds exactly, but it makes the theory tractable and is the natural starting point for extensions. A key consequence concerns prediction: for a zero-mean stationary Gaussian series,

$$E(X_{k+1} \mid X_1, \dots, X_k) = \sum_{i=1}^{k} a_i X_i$$

for some reals a_i, i.e. the best predictor is linear. Given data $X_1, \dots, X_n$, such coefficients can be estimated as in regression, by minimizing $\sum_{t=k+1}^{n} (X_t - a_1 X_{t-1} - \cdots - a_k X_{t-k})^2$.

[Supplement] Proof of linearity of the conditional mean. Let $\mathbf{X} = (X_1, \dots, X_k)^T$, $\Gamma_k = \mathrm{Cov}(\mathbf{X})$ (assume invertible) and $\gamma_k = \mathrm{Cov}(\mathbf{X}, X_{k+1})$. Put $a = \Gamma_k^{-1}\gamma_k$ and $U = X_{k+1} - a^T\mathbf{X}$. Then $\mathrm{Cov}(\mathbf{X}, U) = \gamma_k - \Gamma_k a = 0$. Since $(\mathbf{X}, U)$ is jointly normal (a linear image of a normal vector), zero covariance implies U is independent of $\mathbf{X}$, so $E(U \mid \mathbf{X}) = EU = 0$. Hence $E(X_{k+1} \mid \mathbf{X}) = a^T\mathbf{X} + E(U \mid \mathbf{X}) = a^T\mathbf{X}$. ■

Stationarity is the backbone of time series analysis. Real data are often nonstationary, but transformations (e.g. differencing, leading to the ARIMA models of Chapter 5) can frequently reduce them to stationary ones.

1.3 Selected exercises

Motivation. These textbook exercises practise the basic definitions: the exponential-smoothing recursion, the gap between strict and weak stationarity, Gaussian conditional means, and how autocovariances combine under addition.

Exercise 1.1 (Exponential smoothing recursion)

Let $0 < w < 1$ and let S_t^* be defined by (1.3) from an initial value S_0^*. Show (1.4), i.e.

$$S_t^* = w \sum_{j=0}^{t-1} (1-w)^j x_{t-j} + (1-w)^t S_0^*, \qquad t \ge 1,$$

and deduce that $S_t^* - S_t \to 0$ as $t \to \infty$.

Solution. We argue by induction on t.

Step 1. For $t = 1$, (1.3) gives $S_1^* = wx_1 + (1-w)S_0^*$, which is the claimed formula (the sum has the single term $j = 0$).

Step 2. Let $t \geq 2$ and assume the formula for $t-1$. Substituting it into (1.3),

$$\begin{aligned} S_t^* &= wx_t + (1-w)\Big\{w\sum_{j=0}^{t-2}(1-w)^j x_{t-1-j} + (1-w)^{t-1}S_0^*\Big\} \\ &= wx_t + w\sum_{k=1}^{t-1}(1-w)^k x_{t-k} + (1-w)^t S_0^*, \end{aligned}$$

where $k = j+1$. The term wx_t is the $k = 0$ term of the sum, so $S_t^* = w\sum_{k=0}^{t-1}(1-w)^k x_{t-k} + (1-w)^t S_0^*$.

Step 3. By the definition of S_t, $S_t^* - S_t = (1-w)^t S_0^*$. Since $0 < 1-w < 1$, $(1-w)^t \to 0$, so $S_t^* - S_t \to 0$ for every fixed S_0^*. ■

Exercise 1.3 (Strict versus weak stationarity without independence)

Using examples that are *not* i.i.d., show that
(a) a strictly stationary series need not be weakly stationary;
(b) a weakly stationary series need not be strictly stationary.

Solution.

(a) Let Z be standard Cauchy and $X_t = Z$ for all t. For any $t_1, \ldots, t_k$, h and $x_1, \ldots, x_k$,

$$P(X_{t_1} \leq x_1, \ldots, X_{t_k} \leq x_k) = P(Z \leq \min_i x_i) = P(X_{t_1+h} \leq x_1, \ldots, X_{t_k+h} \leq x_k),$$

so $\{X_t\}$ is strictly stationary. It is not i.i.d., since $X_1 = X_2$ and X_1 is non-degenerate, so X_1, X_2 are dependent. It is not weakly stationary, because $EX_t^2 = EZ^2 = \infty$ violates condition (2) of Definition 1.2.3.

(b) Let $\{X_t\}$ be independent with $X_t \sim N(0,1)$ for odd t, and $P(X_t = 1) = P(X_t = -1) = \frac{1}{2}$ for even t. Every X_t has mean 0 and variance 1, and by independence $\mathrm{Cov}(X_t, X_{t+h}) = \mathbb{1}\{h = 0\}$ for all t. Hence $\{X_t\}$ is weakly stationary (indeed $WN(0,1)$). It is not i.i.d., since the marginal laws differ. It is not strictly stationary: taking $k = 1$, $t_1 = 1$, $h = 1$, $x_1 = \frac{1}{2}$,

$$P(X_1 \leq \tfrac{1}{2}) = \Phi(\tfrac{1}{2}) \approx 0.691 \neq \tfrac{1}{2} = P(X_2 \leq \tfrac{1}{2}),$$

where Φ is the standard normal distribution function. ■

Exercise 1.6 (Gaussian one-step conditional mean)

(a) (Exercise 1.5.) Let (X, Y) be bivariate normal with $\mathrm{Var}(Y) > 0$. Show that $E(X \mid Y) = aY + b$ with $a = \mathrm{Cov}(X,Y)/\mathrm{Var}(Y)$ and $b = EX - aEY$.
(b) Let $\{X_t\}$ be a stationary Gaussian series with mean μ and ACVF γ, $\gamma(0) > 0$. Express $E(X_t \mid X_{t-1})$ and its mean squared error through μ and γ.

Solution.

(a) Put $U = X - aY - b$.

Step 1. (U, Y) is a linear image of the normal vector (X, Y), so every linear combination $sU + tY$ is a linear combination of X, Y and hence univariate normal (possibly degenerate). Therefore (U, Y) has characteristic function $Ee^{i(sU+tY)} = \exp\{i(sEU + tEY) - \frac{1}{2}\mathrm{Var}(sU + tY)\}$.

Step 2. $EU = EX - aEY - b = 0$ and $\operatorname{Cov}(U, Y) = \operatorname{Cov}(X, Y) - a \operatorname{Var}(Y) = 0$. Hence $\operatorname{Var}(sU + tY) = s^2 \operatorname{Var} U + t^2 \operatorname{Var} Y$, and the characteristic function factorizes into a function of s times a function of t. Thus U and Y are independent.

Step 3. Consequently $E(U \mid Y) = EU = 0$, and $E(X \mid Y) = aY + b + E(U \mid Y) = aY + b$.

(b) By Definition 1.2.11, (X_t, X_{t-1}) is bivariate normal with $\operatorname{Var} X_{t-1} = \gamma(0) > 0$ and $\operatorname{Cov}(X_t, X_{t-1}) = \gamma(1)$. Part (a) with $X = X_t$, $Y = X_{t-1}$ gives $a = \gamma(1)/\gamma(0) = \rho(1)$ and $b = \mu - \rho(1)\mu$, so

$$E(X_t \mid X_{t-1}) = \mu + \frac{\gamma(1)}{\gamma(0)} (X_{t-1} - \mu).$$

The prediction error is $U = X_t - \mu - \rho(1)(X_{t-1} - \mu)$, whose mean square is

$$EU^2 = \gamma(0) - 2\rho(1)\gamma(1) + \rho(1)^2\gamma(0) = \gamma(0)(1 - \rho(1)^2) = \gamma(0) - \frac{\gamma(1)^2}{\gamma(0)}.$$

■

Exercise 1.7 (Sum of independent stationary series)

Let $\{X_t\}$ and $\{Y_t\}$ be independent stationary series with ACVFs γ_X, γ_Y. Show that $\{X_t + Y_t\}$ is stationary and find its ACVF.

Solution. Let $Z_t = X_t + Y_t$, $\mu_X = EX_t$, $\mu_Y = EY_t$. We check Definition 1.2.3.

Step 1. $EZ_t = \mu_X + \mu_Y$ for all t.

Step 2. $EZ_t^2 \le 2EX_t^2 + 2EY_t^2 < \infty$.

Step 3. By bilinearity of covariance,

$$\operatorname{Cov}(Z_t, Z_{t+h}) = \operatorname{Cov}(X_t, X_{t+h}) + \operatorname{Cov}(X_t, Y_{t+h}) + \operatorname{Cov}(Y_t, X_{t+h}) + \operatorname{Cov}(Y_t, Y_{t+h}).$$

For any s, u, X_s and Y_u are independent with finite second moments, so $EX_sY_u = EX_s\,EY_u$ and $\operatorname{Cov}(X_s, Y_u) = 0$. Hence

$$\gamma_Z(h) = \operatorname{Cov}(Z_t, Z_{t+h}) = \gamma_X(h) + \gamma_Y(h),$$

which does not depend on t. Thus $\{Z_t\}$ is stationary with ACVF $\gamma_X + \gamma_Y$.

Only $\operatorname{Cov}(X_s, Y_u) = 0$ was used, so uncorrelated series suffice. If $\gamma_X(0) + \gamma_Y(0) > 0$, the ACF is $\rho_Z(h) = \{\gamma_X(0)\rho_X(h) + \gamma_Y(0)\rho_Y(h)\}/\{\gamma_X(0) + \gamma_Y(0)\}$, a variance-weighted average of the two ACFs (with the convention that a term with zero variance is omitted). ■